Notes: Duflo, Greenstone, Pande, and Ryan (QJE, 2013)

Truth-Telling by Third-Party Auditors and the Response of Polluting Firms: Experimental Evidence from India

Contents

1 Big picture

This paper studies whether third-party auditors tell the truth when they are hired and paid by the firms they audit, and whether changing auditor incentives can improve regulatory outcomes. The setting is environmental regulation in Gujarat, India, where high-pollution industrial plants are required to submit annual third-party environmental audits to the Gujarat Pollution Control Board (GPCB).

The status quo audit market had a familiar conflict of interest: plants chose and paid their own auditors. The authors argue that this arrangement gave auditors incentives to underreport pollution so that plants would appear compliant with environmental standards. The paper evaluates a field experiment that changed the structure of the audit market for 473 audit-eligible plants in Ahmedabad and Surat.

The experiment changed auditor incentives through a bundled reform: auditors were randomly assigned to treatment plants, paid from a central pool at a fixed fee, subject to random backchecks by independent technical agencies, and, in the second year, paid partly based on accuracy. The treatment increased truthful reporting by auditors and reduced pollution emissions by plants, especially among the dirtiest plants.

2 Incremental contribution of the paper

1. **Shows that third-party audit markets can be corrupted in equilibrium.** The paper documents that control-group auditors systematically reported pollution readings just below regulatory standards, while independent backchecks showed that true pollution levels were often higher.
2. **Uses a randomized reform of an audit market.** Rather than only documenting bias, the authors experimentally change the market structure: random auditor assignment, fixed central payment, backchecks, and incentive pay.
3. **Links auditor incentives to ultimate regulatory outcomes.** The paper follows the causal chain from auditor incentives to auditor truth-telling and then to plant emissions. Many corruption studies stop at intermediary outcomes; this one measures the environmental endpoint.
4. **Provides direct evidence on third-party auditor conflicts of interest.** The setting is environmental auditing, but the economic problem is shared by financial audits, credit ratings, environmental impact assessments, food safety certification, carbon offsets, and many other audit and certification markets.
5. **Separates reporting behavior from actual pollution.** Backchecks allow the authors to compare audit reports to independently measured pollution, making it possible to detect false compliance and underreporting.
6. **Demonstrates a feasible policy design.** The reform did not require a complete state takeover of monitoring. It changed assignment, payment, verification, and incentives within the existing audit system.

3 Research question and core mechanism

Research question. Can changing the economic incentives of third-party environmental auditors make them report truthfully, and will plants respond by reducing emissions?

Conflict in the status quo. In the original system, audit-eligible plants hire and pay auditors directly. Auditors need plants as clients, so they may shade reports to keep business. Plants benefit when reports show compliance with pollution standards. The regulator receives the audit reports but, absent reliable verification, cannot easily distinguish truthful reports from fabricated reports.

Treatment logic. The reform attempts to make accurate reporting privately valuable for auditors. Random assignment weakens client shopping. Fixed pay from a central pool makes auditor pay independent of plant satisfaction. Backchecks raise the expected cost of lying. Incentive pay in year 2 directly rewards accuracy.

Plant response logic. If plants know that audit reports are more accurate and likely to reach the regulator, then high emissions become more likely to trigger costly regulatory action. Plants therefore have stronger incentives to abate pollution.

4 Institutional setting

4.1 Gujarat pollution regulation

Gujarat is one of India's fastest-growing industrial states and has significant industrial pollution. The Gujarat Pollution Control Board enforces water and air pollution regulations. The board monitors many plants and can impose actions ranging from warning letters and citations to closure notices and utility disconnections.

The study focuses on audit-eligible plants in two heavily industrialized regions, Ahmedabad and Surat. These plants are relatively polluting and are required to submit annual environmental audit reports.

4.2 Environmental audit system

The Gujarat High Court introduced the environmental audit scheme in 1996. Plants with high pollution potential are required to hire certified third-party auditors. Auditors visit plants during three seasons each year, take pollution measurements, and file a final report with both the plant and GPCB.

The system includes formal safeguards, such as auditor certification and a limit on the number of audits per team. However, in practice, plants hire and pay auditors directly, and there is little effective verification in the status quo. Before the experiment, the market price of an audit was often below the estimated cost of collecting and analyzing samples, suggesting that some auditors were not collecting all required readings.

5 Experimental design

5.1 Sample and randomization

The experimental sample contains 473 audit-eligible plants in Ahmedabad and Surat. The authors randomly assign roughly half of the plants to treatment, stratified by region. The treatment group contains 233 plants, and the control group contains 240 plants.

Treatment applies to audit years 2009 and 2010. Control plants remain in the status quo audit market, where they choose and pay auditors directly.

5.2 Treatment components

1. **Auditor assignment.** Treatment plants were randomly assigned an auditor by the research team and GPCB, rather than choosing their own auditor.
2. **Fixed payment from a central pool.** Auditors working for treatment plants were paid from a central pool. The payment was fixed at INR 45,000 in the first year, high enough to cover typical sampling and analysis costs plus a modest margin.
3. **Backchecks.** A random sample of treatment audit readings was verified by independent technical agencies. Backchecks measured the same pollutants at the same locations as the auditor, usually soon after the audit visit.
4. **Incentive pay in year 2.** In the second year, auditor pay was linked to accuracy as measured by backchecks. More accurate auditors received higher per-audit pay.

The treatment was implemented as a package. The authors therefore identify the effect of the bundled market-structure reform, not the separate causal effect of every component. They do, however, use timing evidence to suggest that the year-2 incentive-pay component independently improved reporting.

6 Data

The paper uses four main data sources:

- **Audit reports.** These contain auditor-reported readings for water pollutants such as biochemical oxygen demand (BOD), chemical oxygen demand (COD), total dissolved solids (TDS), total suspended solids (TSS), and ammoniacal nitrogen (NH3-N), and air pollutants such as sulfur dioxide (SO2), nitrogen oxides (NOx), and suspended particulate matter (SPM).
- **Backchecks.** Independent technical agencies collected readings for the same pollutants, allowing a direct comparison between audit reports and true measured pollution.
- **Endline pollution survey.** Roughly six months after the final audit visits, independent agencies measured pollution for both treatment and control plants.
- **GPCB administrative records.** These include inspections and regulatory actions, allowing the authors to study how penalties respond to pollution violations.

7 Main empirical specifications

7.1 Status quo reporting in the control group

To compare audit reports and backchecks in the control group, the paper estimates regressions of the form:

$$1\{\text{Compliant}\}_{ij} = \alpha_1 1\{\text{AuditReport}\}_{ij} + \gamma_r + \varepsilon_{ij}.$$

The dependent variable is either narrow compliance, meaning the reading is between 75% and 100% of the regulatory standard, or compliance, meaning the reading is at or below the standard. The coefficient on the audit-report indicator measures whether auditors report compliance more often than independent backchecks do.

7.2 Treatment effect on false compliance

To compare treatment and control reporting, the paper estimates:

$$1\{\text{Compliant}\}_{ij} = \alpha_1\big(1\{\text{AuditReport}\}_{ij} \times T_j\big) + \alpha_2 1\{\text{AuditReport}\}_{ij} + \alpha_3 T_j + \gamma_r + \varepsilon_{ij}.$$

Here T_j is the treatment indicator. The interaction coefficient measures how the treatment changes the gap between auditor reports and backchecks.

7.3 Treatment effect on reported pollution levels

The paper also studies standardized pollution readings in audit reports:

$$y_{ijt} = \beta T_j + \gamma_r + \gamma_t + \varepsilon_{ijt}.$$

Pollutants are standardized relative to the distribution of backcheck readings. Some specifications include auditor fixed effects, which compare the same auditor working in treatment and control plants.

7.4 Treatment effect on endline plant emissions

To estimate the plant pollution response, the paper regresses endline pollution outcomes on treatment and region fixed effects. It also estimates quantile treatment effects:

$$Q_{y_{ij}|X_j}(\tau) = \beta_\tau T_j + \gamma_r + \varepsilon_{ij}.$$

This shows where in the pollution distribution emissions reductions occur.

8 Main findings

8.1 Finding 1: Status quo audits are corrupted

Control-group audit reports show substantial bunching just below pollution standards. For SPM, 73% of audit readings fall between 75% and 100% of the standard, compared with only 19% of backcheck readings. Audit reports make plants look compliant far more often than the independent measurements do.

Across all pollutants, audit reports in the control group are 27 percentage points more likely than backchecks to show narrow compliance and 28.8 percentage points more likely to show compliance. This indicates systematic false reporting, not merely random measurement error.

8.2 Finding 2: The treatment increases truthful reporting

The treatment sharply reduces false compliance. Across all pollutants, the treatment reduces the audit-report versus backcheck gap in compliance by 23.4 percentage points. In other words, treatment audits are much less likely to falsely report that a plant is compliant.

The treatment also increases reported pollution levels. Audit reports for treatment plants show pollution levels about 0.10 to 0.13 standard deviations higher than audit reports for control plants. When comparing audit readings directly to backchecks in the midline sample, the treatment closes roughly 70% of the underreporting gap.

8.3 Finding 3: Auditor fixed effects do not explain the result

Specifications with auditor fixed effects show that the same auditors report differently when operating under the treatment and control systems. This is important because it rules out the explanation that treatment plants simply received better auditors. The treatment changed auditor behavior.

8.4 Finding 4: Incentive pay appears to improve reporting

In year 2, incentive pay was introduced. Reported pollution in treatment plants jumps up at the start of the second year relative to the control group, interrupting a downward trend in reported treatment pollution. This nonexperimental timing evidence suggests that explicit financial incentives for accuracy improved reporting on top of the other treatment components.

8.5 Finding 5: Treatment plants reduce pollution

At the endline, pollution in treatment plants is 0.211 standard deviations lower than in control plants across all pollutants. The reduction is driven mainly by water pollutants and is concentrated among plants with the highest pollution readings. Compliance rates do not increase significantly on average, indicating that reductions come from the right tail of the pollution distribution rather than from marginal plants just above the standard.

8.6 Finding 6: Regulatory action is strongest for extreme violations

GPCB penalties become more severe as the degree of violation rises. The harshest actions, such as closure notices and utility disconnections, are concentrated among plants with very high pollution readings. This penalty structure helps explain why the treatment induced the largest emission reductions among the dirtiest plants.

9 How to interpret the treatment channels

The treatment combines multiple components, so the main estimates identify the bundled reform. The paper offers several arguments about the channels:

- **Random assignment** weakens client shopping and the auditor's need to please the plant to keep future business.
- **Fixed payment from a central pool** reduces direct dependence of auditor pay on plant satisfaction and ensures that auditors can afford to collect real samples.
- **Backchecks** raise the probability that false reporting is detected.
- **High enough payment plus monitoring** can create an efficiency-wage channel: inaccurate reporting risks losing future auditing income.
- **Incentive pay** explicitly rewards accuracy in year 2 and appears to have an independent effect.

The within-auditor estimates rule out purely auditor-level explanations, such as better auditors selecting into treatment or a general income effect that would improve an auditor's reporting in both treatment and control plants.

10 Extracted figures and tables from the paper

10.1 Table I: Submission of Audit Reports

Read. Table I shows that audit submission rates are similar between treatment and control plants in both years. In 2009, 70% of treatment plants and 74% of control plants submitted audit reports. In 2010, 70% of treatment plants and 64% of control plants submitted reports.

Economic message. The treatment did not primarily work by increasing the probability that plants filed reports. The main treatment margin is the content and truthfulness of audit reports conditional on reporting.

10.2 Table II: Audit Treatment Covariate Balance

Read. Table II checks balance for plants that submitted an audit report in either year. Treatment and control plants are similar across plant characteristics and prior regulatory interactions. The only notable imbalance is that treatment plants are less likely to have a bag filter installed and somewhat more likely to have had an air citation before the study.

Economic message. Randomization generally worked. Baseline similarity helps interpret subsequent differences in reporting and pollution as treatment effects rather than preexisting differences.

TABLE I

SUBMISSION OF AUDIT REPORTS

	(1) Treatment	(2) Control	(3) Difference
Panel A: 2009			
Audit submitted	163	177	
Total plants	233	240	
Share submitted	0.70	0.74	−0.038
			(0.041)
Panel B: 2010			
Audit submitted	164	153	
Total plants	233	240	
Share submitted	0.70	0.64	0.066
			(0.043)

Notes. The table reports on the number of audit reports submitted to the regulator for plants in the audit sample over the two years of the experiment. Column (3) shows differences between treatment and control group submission rates with standard errors in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$ throughout (neither difference here is significant at these levels).

Table I: Submission of Audit Reports

TABLE II

AUDIT TREATMENT COVARIATE BALANCE

	(1) Treatment	(2) Control	(3) Difference
Panel A: Plant characteristics			
Capital investment INR 50 m to 100 m (= 1)	0.092	0.14	−0.051
	[0.29]	[0.35]	(0.033)
Located in industrial estate (= 1)	0.57	0.53	0.042
	[0.50]	[0.50]	(0.051)
Textiles (= 1)	0.88	0.93	−0.030
	[0.33]	[0.26]	(0.025)
Effluent to common treatment (= 1)	0.41	0.35	0.078
	[0.49]	[0.48]	(0.049)
Wastewater generated (kl/day)	420.5	394.6	35.4
	[315.9]	[323.4]	(31.6)
Lignite used as fuel (= 1)	0.71	0.77	−0.024
	[0.45]	[0.42]	(0.029)
Diesel used as fuel (= 1)	0.29	0.25	0.038
	[0.45]	[0.43]	(0.046)
Air emissions from flue gas (= 1)	0.85	0.87	−0.0095
	[0.35]	[0.33]	(0.016)
Air emissions from boiler (= 1)	0.93	0.92	0.026
	[0.26]	[0.27]	(0.027)
Bag filter installed (= 1)	0.24	0.34	−0.10**
	[0.43]	[0.47]	(0.046)
Cyclone installed (= 1)	0.087	0.079	0.0010
	[0.28]	[0.27]	(0.027)
Scrubber installed (= 1)	0.41	0.41	−0.018
	[0.49]	[0.49]	(0.050)
Panel B: Regulatory interactions in year prior to study			
Whether audit submitted (= 1)	0.82	0.81	0.022
	[0.38]	[0.39]	(0.038)
Any equipment mandated (= 1)	0.42	0.49	−0.047
	[0.50]	[0.50]	(0.047)
Any inspection conducted (= 1)	0.79	0.78	0.016
	[0.41]	[0.42]	(0.042)
Any citation issued (= 1)	0.28	0.24	0.035
	[0.45]	[0.43]	(0.045)
Any water citation issued (= 1)	0.12	0.12	−0.0031
	[0.33]	[0.33]	(0.034)
Any air citation issued (= 1)	0.027	0.0052	0.021*
	[0.16]	[0.072]	(0.013)
Any utility disconnection (= 1)	0.098	0.094	0.0029
	[0.30]	[0.29]	(0.031)
Any bank guarantee posted (= 1)	0.033	0.026	0.0045
	[0.18]	[0.16]	(0.017)
Observations	184	191	

Notes. The sample includes firms in the audit sample that submitted an audit report in either year; the balance for all audit sample firms, discussed in the text, is similar. Columns (1) and (2) show means with standard deviations in brackets. Column (3) shows the coefficient on treatment from regressions of each characteristic on treatment and region fixed effects. 50 INR ≈ US$1. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table II: Audit Treatment Covariate Balance

10.3 Figure I: Readings for Suspended Particulate Matter (SPM, mg/Nm3), Midline

Read. Figure I compares audit and backcheck distributions for SPM in control and treatment plants. In the control group, audit readings bunch just below the standard, but backchecks are widely dispersed and often above the standard. In the treatment group, the audit distribution is much closer to the backcheck distribution.

Economic message. This figure is the clearest visual evidence of status quo false reporting and treatment-induced truth-telling. The treatment reduces the fabricated mass just below the standard.

10.4 Table III: Compliance in Audits Relative to Backchecks, Control Group Only

Read. Table III uses control plants only. Audit reports are 27 percentage points more likely than backchecks to show narrow compliance and 28.8 percentage points more likely to show compliance.

Economic message. Under the status quo, auditors report plants as compliant much more often than independent measurements do. This confirms the corruption of the baseline audit market.

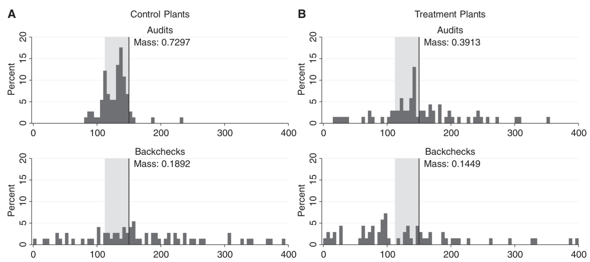

FIGURE I

Readings for Suspended Particulate Matter (SPM, mg/Nm^3), Midline

The figure shows distributions of pollutant concentrations for SPM in boiler-stack samples taken during the midline survey. Panel A shows the distributions of readings at control plants from audits and backchecks, respectively, and Panel B readings at treatment plants from the same two sources. The regulatory maximum concentration limit of 150 mg/Nm^3 for SPM is marked with a vertical line, and the area between 75% and 100% of the limit is shaded in gray.

Figure I: Readings for Suspended Particulate Matter (SPM, mg/Nm3), Midline

TABLE III

COMPLIANCE IN AUDITS RELATIVE TO BACKCHECKS, CONTROL GROUP ONLY

	(1) All pollutants	(2) Water pollutants	(3) Air pollutants
Panel A: Dependent variable: Narrow compliance (dummy for pollutant between 75% and 100% of regulatory standard)			
Audit report (= 1)	0.270***	0.297***	0.230***
	(0.025)	(0.034)	(0.033)
Control mean in backchecks	0.097	0.110	0.077
Panel B: Dependent variable: Compliance (dummy for pollutant at or below regulatory standard)			
Audit report (= 1)	0.288***	0.273***	0.311***
	(0.023)	(0.033)	(0.032)
Control mean in backchecks	0.557	0.538	0.586
Observations	1132	688	444

Notes. Regressions include region fixed effects. "Audit report" is a dummy for a pollutant reading reported in an audit, as opposed to reported in a backcheck, which is the omitted category. Sample of matched plant-by-pollutant pairs from audit reports submitted to the regulator in the control group only and corresponding backchecks from the midline survey. Pollution samples from final-stage effluent outlet for water and boiler stack for air. Pollutants included are *Water* = {NH_3-N, BOD, COD, TDS, TSS} and *Air* = {SO_2, NO_x, SPM} with $All = Water \cup Air$. Standard errors clustered at the plant level in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table III: Compliance in Audits Relative to Backchecks, Control Group Only

10.5 Table IV: Compliance in Audits Relative to Backchecks by Treatment Status

Read. Table IV shows that treatment auditors are much less likely to falsely report compliance. Across all pollutants, the treatment reduces false compliance by 23.4 percentage points and narrow-compliance bunching by 18.5 percentage points.

Economic message. The audit market reform changes reporting at the regulatory threshold. This is especially policy-relevant because compliance with the standard is the decision margin that regulators observe.

10.6 Figure II: Audit Treatment Effect in Density Bins, All Pollutants

Read. Figure II plots treatment effects across bins of standardized pollution relative to the regulatory limit. The treatment sharply reduces reported mass just below the standard and increases mass above the standard.

Economic message. Treatment auditors are no longer concentrating reports in the safe zone below the standard. The figure directly visualizes the reduction in strategic misreporting.

TABLE IV

COMPLIANCE IN AUDITS RELATIVE TO BACKCHECKS BY TREATMENT STATUS

	(1) All pollutants	(2) Water pollutants	(3) Air pollutants
Panel A: Dependent variable: Narrow compliance (dummy for pollutant between 75% and 100% of regulatory standard)			
Audit report × Treatment group	−0.185***	−0.212***	−0.143***
	(0.034)	(0.044)	(0.046)
Audit report (= 1)	0.270***	0.297***	0.230***
	(0.025)	(0.034)	(0.033)
Treatment group (= 1)	−0.0034	−0.013	0.011
	(0.0176)	(0.025)	(0.024)
Control mean in backchecks	0.097	0.110	0.077
Panel B: Dependent variable: Compliance (dummy for pollutant at or below regulatory standard)			
Audit report × Treatment group	−0.234***	−0.166***	−0.345***
	(0.039)	(0.050)	(0.056)
Audit report (= 1)	0.288***	0.273***	0.311***
	(0.023)	(0.033)	(0.032)
Treatment group (= 1)	0.058*	0.0075	0.145***
	(0.034)	(0.0477)	(0.041)
Control mean in backchecks	0.557	0.538	0.586
Observations	2236	1378	858

Notes. Regressions include region fixed effects. "Treatment group" is a dummy equal to 1 for plants where auditors were randomly assigned, paid a fixed rate from a common pool and subject to backchecks. "Audit report" is a dummy for a pollutant reading reported in an audit, as opposed to reported in a backcheck, which is the omitted category. Sample of matched plant-by-pollutant pairs from audit reports submitted to the regulator and corresponding backchecks from the midline survey, in both the treatment and control groups. Pollution samples from final-stage effluent outlet for water and boiler stack for air. Pollutants included are $Water = \{NH_3\text{-N, BOD, COD, TDS, TSS}\}$ and $Air = \{SO_2, NO_x, \text{SPM}\}$ with $All = Water \cup Air$. Standard errors clustered at the plant level in parentheses. *$p < .10$, **$p < .05$, ***$p < .01$.

Table IV: Compliance in Audits Relative to Backchecks by Treatment Status

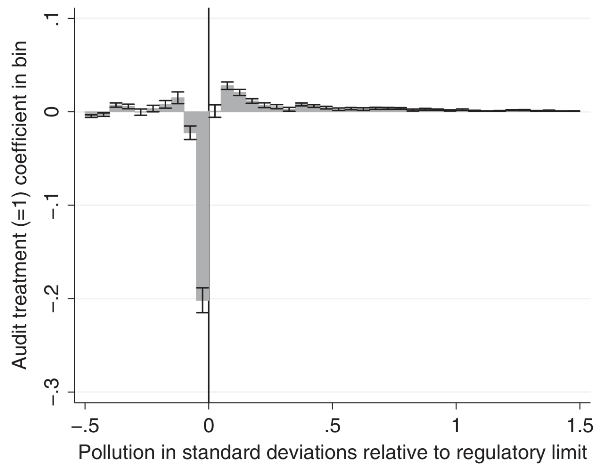

FIGURE II

Audit Treatment Effect in Density Bins, All Pollutants

The figure reports point estimates and standard errors from 40 OLS regressions where the dependent variables are indicators for a pollutant reading being within a given density bin and the independent variables are region fixed effects and the audit treatment dummy. All pollutants are included with $Water = \{NH_3\text{-N, BOD, COD, TDS, TSS}\}$ and $Air = \{SO_2, NO_x, \text{SPM}\}$, and $All = Water \cup Air$. Pollutants are standardized by subtracting the regulatory standard for each pollutant and dividing by the standard deviation of that pollutant in backchecks. Density bins are 0.05 standard deviations wide.

Figure II: Audit Treatment Effect in Density Bins, All Pollutants

10.7 Table V: Audit Treatment Effects on Auditor Reporting

Read. Table V reports treatment effects on compliance, reported pollution levels, and audit-backcheck differences. Treatment reports are less likely to show compliance and report higher pollution levels. The results remain similar with auditor fixed effects.

Economic message. The treatment affects the behavior of the same auditors working in different market structures. This strongly supports an incentives interpretation rather than an auditor-selection story.

10.8 Figure III: Time Series of Audit Reports by Treatment Status, All Pollutants

Read. Figure III shows mean reported pollution over the two audit years and three reporting seasons. Treatment reports are higher than control reports, but decline through year 1. At the start of year 2, when incentive pay begins, treatment reports jump upward.

Economic message. The timing supports an independent role for incentive pay. The decline in treatment reports over time also anticipates the finding that plants reduce actual pollution.

TABLE V

AUDIT TREATMENT EFFECTS ON AUDITOR REPORTING

	(1)	(2)	(3)	(4)	(5)	(6)
	All pollutants		Water pollutants		Air pollutants	
Panel A: Dependent variable: Compliance (dummy for pollutant in audit report at or below regulatory standard)						
Treatment group (= 1)	−0.151***	−0.180***	−0.172***	−0.188***	−0.108***	−0.149***
	(0.015)	(0.022)	(0.021)	(0.030)	(0.016)	(0.021)
Auditor fixed effects	No	Yes	No	Yes	No	Yes
Control mean	0.828	0.828	0.807	0.807	0.863	0.863
Observations	13170	13170	8373	8373	4797	4797
Panel B: Dependent variable: Level of pollutant in audit report (standard deviations relative to backcheck mean)						
Treatment group (= 1)	0.103***	0.131***	0.117**	0.131**	0.0853***	0.132***
	(0.035)	(0.038)	(0.053)	(0.059)	(0.0214)	(0.022)
Auditor fixed effects	No	Yes	No	Yes	No	Yes
Control mean	−0.291	−0.291	−0.350	−0.350	−0.194	−0.194
Observations	13170	13170	8373	8373	4797	4797
Panel C: Dependent variable: Level of pollutant in audit report minus level of pollution in backcheck (standard deviations)						
Treatment group (= 1)	0.210***	0.153	0.152	0.156	0.312***	0.166*
	(0.073)	(0.099)	(0.102)	(0.138)	(0.083)	(0.095)
Auditor fixed effects	No	Yes	No	Yes	No	Yes
Control mean	−0.304	−0.304	−0.354	−0.354	−0.225	−0.225
Observations	1118	1118	689	689	429	429

Notes. Regressions include region fixed effects in all panels and year fixed effects in Panels A and B only. Pollution samples are from the final-stage effluent outlet for water and boiler stack for air. Pollutants included are *Water* = {NH_3-N, BOD, COD, TDS, TSS} and *Air* = {SO_2, NO_x, SPM} with *All* = *Water* ∪ *Air*. Panels A and B use the full sample of audit reports that reached the regulator over the two years of the experiment. The outcome variable is a dummy for whether a pollutant was reported compliant in audit reports, in Panel A, and the level of pollution reports in Panel B. Panel C uses the midline survey sample with pollution in audit reports less pollution in backchecks as the outcome. Standard errors clustered at the plant level are in parentheses. $^*p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table V: Audit Treatment Effects on Auditor Reporting

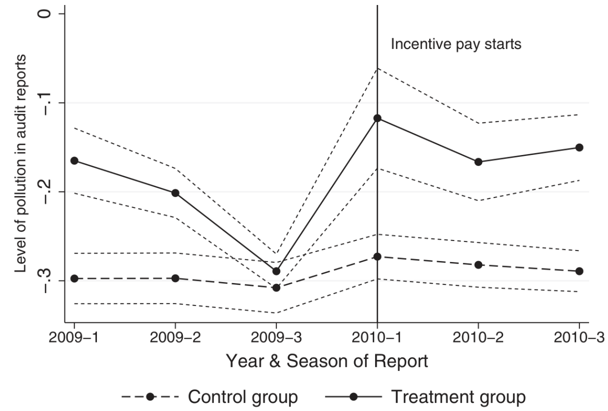

FIGURE III

Time Series of Audit Reports by Treatment Status, All Pollutants

The figure reports the mean standardized pollution level reported in audits by time and treatment status. Time is divided into two years and the three seasons of the year in which auditors are required to monitor pollution. All pollutants are included with *Water* = {NH_3-N, BOD, COD, TDS, TSS} and *Air* = {SO_2, NO_x, SPM} and *All* = *Water* ∪ *Air*. Pollutants are standardized by subtracting the mean for each pollutant and dividing by the standard deviation, where both statistics are calculated from backchecks of that pollutant. The dotted lines around the mean reports in each group give 95% confidence intervals for the mean using standard errors clustered at the plant level.

Figure III: Time Series of Audit Reports by Treatment Status, All Pollutants

10.9 Table VI: Incentive Pay from Treatment Effect over Time

Read. Table VI formalizes the time-series evidence. The interaction between incentive pay and treatment is positive and significant: reported pollution in treatment plants rises by 0.257 standard deviations at the start of year 2.

Economic message. Explicit accuracy incentives appear to improve reporting beyond the general effects of assignment, central payment, and backchecks.

10.10 Table VII: Endline Pollutant Concentrations on Treatment Status

Read. Table VII estimates the effect of audit treatment on actual endline pollution. Across all pollutants, treatment plants have pollution 0.211 standard deviations lower than control plants. The reduction is larger for water pollutants, at 0.300 standard deviations.

Economic message. Improved reporting changes plant behavior. The treatment does not merely clean up paperwork; it leads to lower pollution emissions.

TABLE VI

INCENTIVE PAY FROM TREATMENT EFFECT OVER TIME

	(1)	(2)
	Level of pollutant in audit report, all pollutants (standard deviations)	
Treatment group (= 1)	0.103***	0.218***
	(0.035)	(0.063)
Incentive pay (year = 2010)	0.051**	0.002
	(0.027)	(0.033)
Incentive pay × treatment		0.257***
		(0.092)
Years (fractional) from Jan. 1, 2009		0.029
		(0.023)
Years (fractional) × treatment		−0.220***
		(0.068)
Region fixed effects	Yes	Yes
Observations	13,166	13,166

Notes. Regressions include region fixed effects. Pollution samples from final-stage effluent outlet for water and boiler stack for air. Sample of all audit reports to the regulator over the two years of the experiment. Pollutants included are $Water = \{NH_3\text{-N}, BOD, COD, TDS, TSS\}$ and $Air = \{SO_2, NO_x, SPM\}$ with $All = Water \cup Air$. Standard errors clustered at the plant level in parentheses. *$p < .10$, **$p < .05$, ***$p < .01$.

Table VI: Incentive Pay from Treatment Effect over Time

TABLE VII

ENDLINE POLLUTANT CONCENTRATIONS ON TREATMENT STATUS

	(1) All pollutants	(2) Water pollutants	(3) Air pollutants
Panel A: Dependent variable: Level of pollutant in endline survey, all pollutants (standard deviations relative to backcheck mean)			
Audit treatment assigned (= 1)	−0.211**	−0.300*	−0.053
	(0.099)	(0.159)	(0.057)
Control mean	0.076	0.114	0.022
Observations	1439	860	579
Panel B: Dependent variable: Compliance (dummy for pollutant in endline survey at or below regulatory standard)			
Audit treatment assigned (=1)	0.027	0.039	0.002
	(0.027)	(0.039)	(0.028)
Control mean	0.573	0.516	0.656
Observations	1,439	860	579

Notes. Regressions include region fixed effects. Pollution samples from final-stage effluent outlet for water and boiler stack for air. Pollutants included are $Water = \{NH_3\text{-N}, BOD, COD, TDS, TSS\}$ and $Air = \{SO_2, NO_x, SPM\}$ with $All = Water \cup Air$. Endline survey data in the audit sample of plants not subject to the cross-cut experimental treatment. Standard errors clustered at the plant level in parentheses. *$p < .10$, **$p < .05$, ***$p < .01$.

Table VII: Endline Pollutant Concentrations on Treatment Status

10.11 Figure IV: Quantile Treatment Effects of Audit on Endline Pollution

Read. Figure IV shows quantile treatment effects across the pollution distribution. The effects are close to zero for low and middle quantiles but become sharply negative in the upper tail.

Economic message. Pollution reductions are concentrated among the dirtiest plants. These are precisely the plants most likely to face serious regulatory penalties once reports become accurate.

10.12 Figure V: Regulatory Actions by Degree of Violation

Read. Figure V shows that GPCB actions become more severe as pollution readings exceed the standard by larger multiples. Extreme violations are much more likely to receive closure notices or utility disconnection orders.

Economic message. The regulator's penalty structure explains why the treatment has its largest pollution effects in the right tail. The dirtiest plants have the most to lose from accurate reporting.

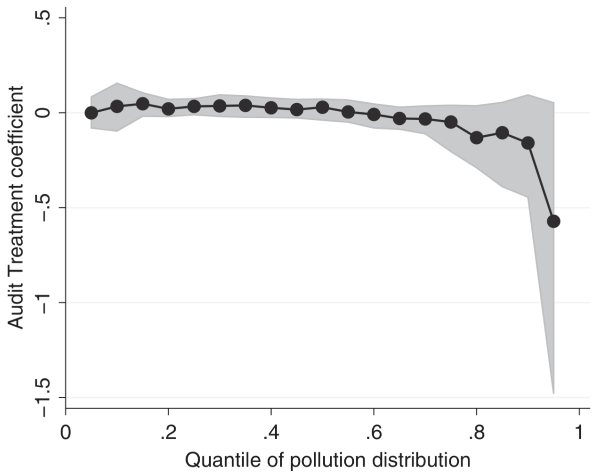

Figure IV

Quantile Treatment Effects of Audit on Endline Pollution

The figure reports estimates from quantile regressions of standardized endline pollution for all pollutants on a dummy for audit treatment assignment and region fixed effects in the audit sample of plants not subject to the cross-cut experimental treatment, analogous to the OLS specifications in Table VII. The quantiles are from 0.05 quantile to the 0.95 quantile at 0.05-quantile intervals. The gray area shows 95% confidence intervals for the treatment coefficient at each quantile from a cluster-bootstrap with replacement at the plant level with 200 replications.

Figure IV: Quantile Treatment Effects of Audit on Endline Pollution

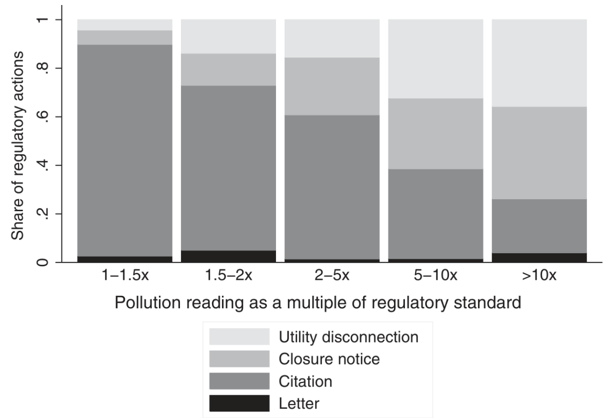

Figure V

Regulatory Actions by Degree of Violation

The figure reports the regulatory responses to pollution readings measured at different levels of noncompliance during regulatory inspections for audit sample plants over the three years beginning one year prior to the study. Pollutant readings are shown in bins of readings at specified multiples above the regulatory standard. The bars indicate the type of regulatory action taken in response to a given reading. Actions increase in severity from bottom (dark bars) to top (light bars): a *letter* is official but not legal correspondence to the firm noting the violation and possibly threatening action, a *citation* is a legal regulatory notice requiring a response from the firm, a *closure notice* is a warning that the plant will be closed unless a violation is remedied, and a *disconnection* is an order to the utility that a plant's power be disconnected. All of these actions were coded based on complete administrative records of plant interactions with the regulator. Going left to right across the bars, the number of violating plants (actions) used to calculate the action shares at each degree of violation are 153 (305), 102 (159), 141 (178), 70 (120), and 72 (126).

Figure V: Regulatory Actions by Degree of Violation

10.13 Appendix Table A.1: Pollutant Descriptions - Panel A: Water Pollutants

Read. Appendix Table A.1 defines the water pollutants used in the paper: BOD, COD, TDS, TSS, and NH3-N. These pollutants measure oxygen demand, dissolved and suspended solids, and nitrogen-related toxicity.

Economic message. The pollution outcomes are not abstract indices. They measure concrete environmental damages, including oxygen depletion, mineralization, turbidity, toxicity to aquatic life, and potential human health effects.

10.14 Appendix Table A.1: Pollutant Descriptions - Panel B: Air Pollutants

Read. Appendix Table A.1 also defines the air pollutants used in the paper: SO2, NOx, and SPM. These pollutants are linked to respiratory and cardiovascular harm and to fine particle formation.

Economic message. The treatment's emissions reductions matter because these pollutants have well-known health and environmental consequences. The appendix clarifies why the audited parameters are socially important.

APPENDIX TABLE A.1

Pollutant Descriptions

Pollutant	Description
Panel A: Water pollutants	
Biochemical oxygen demand (BOD)	A measure of the amount of dissolved oxygen consumed by microscopic organisms in a confined sample of water. The BOD and volume of an effluent determine the oxygen demand that will be imposed on receiving waters (Boyd 2000). The demand for oxygen from effluent may deplete available molecular oxygen, precluding other biological processes, such as marine plants or life, that require oxygen (Waite 1984).
Chemical oxygen demand (COD)	A measure of the oxygen demand of the organic matter in a sample as determined by oxidation of the organic matter with potassium dichromate and sulfuric acid. Often used as a proxy for BOD in determining the oxygen demand of effluent.
Total dissolved solids (TDS)	Primarily inorganic substances dissolved in water, including calcium, magnesium, sodium, potassium, iron, zinc, copper, manganese. Water with high dissolved solids is said to be mineralized and decreases the survival of plant and animal life, degrades the taste of water, corrodes plumbing, and limits use of water for irrigation (IHD-WHO Working Group 1978; Boyd 2000). Depending on the composition of solids, TDS may have adverse health effects on people with cardiac disease or high blood pressure.
Total suspended solids (TSS)	Organic and inorganic or mineral particles too large to be dissolved but small enough to remain suspended against gravity in an effluent (Boyd 2000). Contribute to turbidity and color of water and proxy for adverse effects from individual solid components.
Ammoniacal nitrogen (NH_3-N)	The nitrogen contained in unionized ammonia and ammonium. Though nitrogen is a vital nutrient, some forms of ammonia nitrogen are toxic to aquatic life (Boyd 2000). The toxicity of ammonia nitrogen increases with decreasing dissolved oxygen concentrations.

Appendix Table A.1: Pollutant Descriptions - Panel A: Water Pollutants

APPENDIX TABLE A.1

(continued)

Pollutant	Description
Panel B: Air pollutants	
Sulfur dioxide (SO_2)	A reactive oxide of sulfur. Short-term exposure has been linked to adverse respiratory effects particularly damaging for asthmatics. SO_2 also contributes to formation of fine particles (World Health Organization 2006).
Nitrogen oxides (NO_x)	A group of reactive gases including nitrous acid, nitric acid, and NO_2. Nitrogen oxides are toxic at high concentrations and contribute to formation of ozone and fine particles, which are detrimental to health (World Health Organization 2006).
Suspended particulate matter (SPM)	A mixture of small particles and liquid droplets with a number of components, including acids, organic chemicals, metals, and soil or dust (U.S. Environmental Protection Agency 2010). Particulate matter affects respiratory and cardiovascular health and has been shown to increase infant mortality and shorten life spans (World Health Organization 2006; Currie and Walker 2011; Chen et al. 2013).

Notes. The table describes the pollutants used throughout the analysis of both auditor reporting and plant pollution emissions. Auditor incentive pay during year 2 of the experiment was based on these pollutants and the water parameter pH. We omit pH from the analysis because environmental damages from pH are not monotonic in the reading—both high (alkanine) and low (acidic) readings can be bad—which makes results for pH incomparable to results for other pollutants. For this same reason, the standard for pH is a range, rather than a maximum.

Appendix Table A.1: Pollutant Descriptions - Panel B: Air Pollutants

11 How to read the evidence

11.1 Why backchecks are central

The backchecks create an independent measure of the truth. Without them, audit bunching below the standard could be misread as evidence that plants precisely target compliance. With backchecks, it is clear that much of the bunching is false reporting.

11.2 Why auditor fixed effects matter

Auditor fixed effects compare the same audit firm across treatment and control plants. The fact that treatment effects survive these fixed effects implies that the treatment changed incentives and behavior, not just the mix of auditors.

11.3 Why endline pollution matters

Improved audit reports are only valuable if regulation uses them or plants believe regulators will use them. The endline results show that plants responded by reducing emissions, especially when the expected penalty from truthful reporting was highest.

12 Policy implications

The paper suggests that third-party auditing can work if auditors are not financially dependent on the audited party and if there is credible verification. The most direct policy lesson is that audit markets should be designed so that the first-order incentive is to tell the truth. Random assignment, central payment, verification, and accuracy incentives are practical ways to move in that direction.

The findings also suggest that information reforms can have real environmental effects. Regulators often lack accurate information; improving the truthfulness of monitoring can make existing regulation more effective without changing formal standards.

13 Limitations and open questions

- **Bundled treatment.** The experiment identifies the effect of the reform package, not the clean separate effect of every component.
- **External validity.** The setting is environmental auditing in Gujarat. Other third-party audit markets may differ in complexity, reputation, concentration, and political economy.
- **Long-run sustainability.** The study covers two years. In the long run, auditors, plants, and backcheckers might adapt, possibly by forming new collusive arrangements.
- **Regulatory follow-through.** The plant response depends on the expectation that the regulator will act on truthful reports. In a setting with no enforcement capacity, audit reform alone might have weaker effects.
- **Cost-benefit uncertainty.** The paper's tentative cost-benefit calculation suggests net benefits, but it depends on assumptions about marginal damages and abatement costs.

14 Short summary

Duflo, Greenstone, Pande, and Ryan (2013) evaluate a two-year randomized reform of the environmental audit market in Gujarat, India. In the status quo, auditors hired and paid by plants systematically underreported pollution, often bunching reports just below regulatory standards. The treatment randomly assigned auditors, paid them from a central pool, backchecked their readings, and later rewarded accuracy. The reform increased truthful reporting and sharply reduced false compliance. Plants responded by lowering pollution, with the largest reductions among the highest polluters. The paper shows that changing incentives in third-party audit markets can improve information quality and make regulation more effective.

15 Conclusion

The paper's central lesson is that the design of auditor incentives matters. When auditors are chosen and paid by the firms they audit, third-party verification can become a rubber stamp. When auditors are assigned independently, paid in a way that does not depend on pleasing the client, monitored, and rewarded for accuracy, they report more truthfully. In this experiment, truthful reporting translated into lower pollution. The paper therefore provides unusually complete evidence linking audit-market design to regulatory information and real social outcomes.